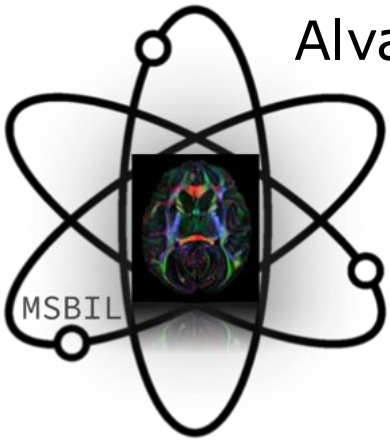




Ferrets as a Preclinical Aging Model: Investigating behavior



Alvaro Cruz Perez, member of Multi-scale Brain Imaging Lab led
by Dr. Elizabeth Hutchinson



Background

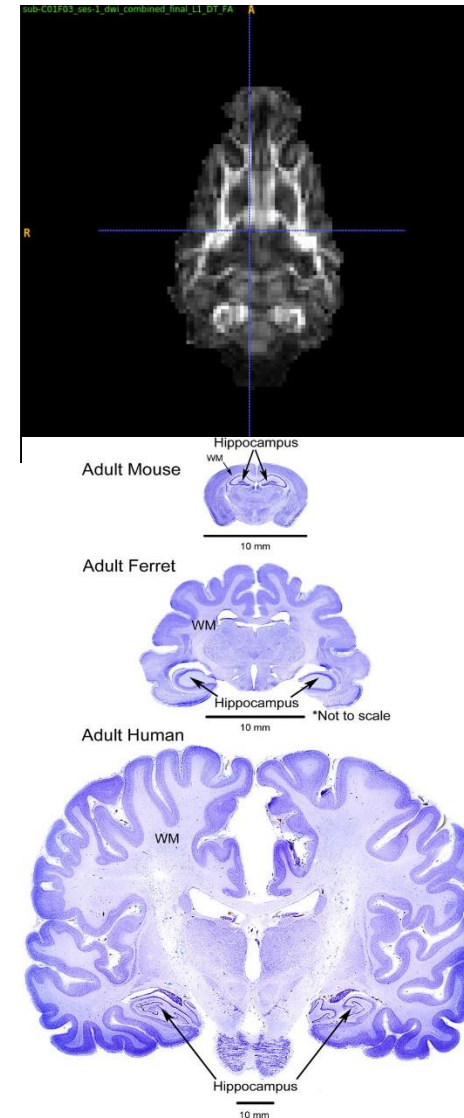
Relationship between behavior and brain structure using **MRI** and **behavioral assessments**

Human behavior connection

- Physical activity (Kraft, 2012)
- Social interaction (Kelly et al., 2017)
- Sleep patterns (Bah et al., 2019)

Why ferrets?

- Geriatric by 3-4 years old (Bakthavatchalu et al., 2016)
- Gyrencephalic brains (used on TBI and neurodevelopment studies) (Schwerin et al., 2017)



(Schwerin et al., 2017)

Ferret behaviors



Burrowing



Social interaction

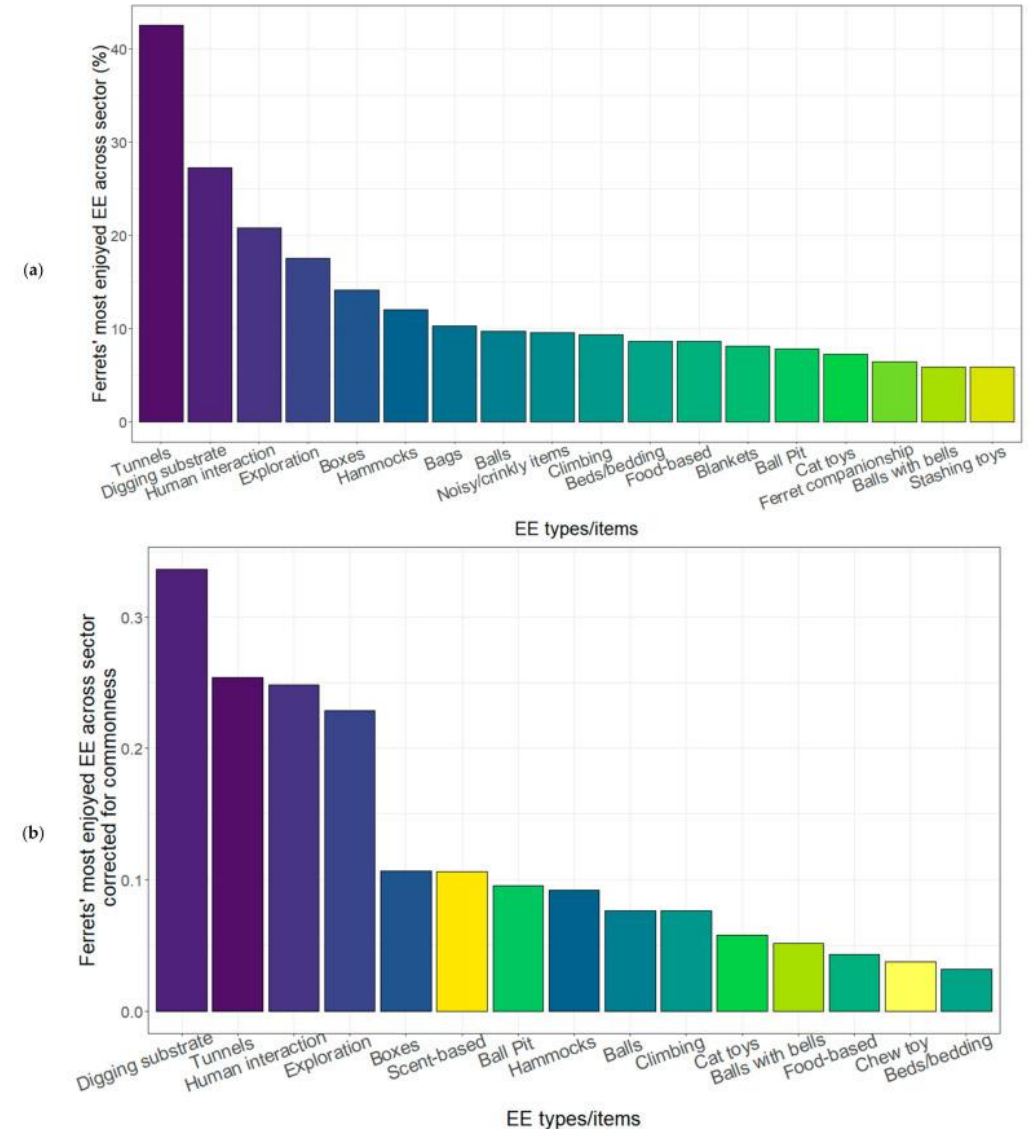


Sleep patterns



Burrowing is a naturalistic behavior preferred by ferrets

- Burrowing time/enrichment time
- Data collection:
 - **10 minutes** videos
 - Videos from **2023** (old ferrets)
 - One video of each **old** ferret from **2024**
 - One video of each **young** ferret from **2024**



(Köbrunner et al., 2020)



Subjects:

- 4 aged female ferrets (~ 4 years old)
- 4 young female ferrets (10 months old)
- 4 young male ferrets (10 months old)

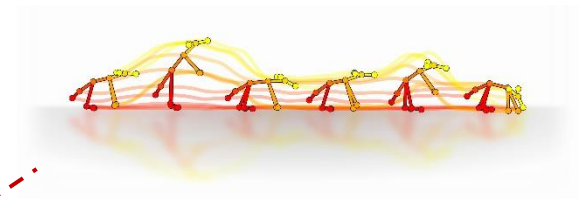


Video recordings

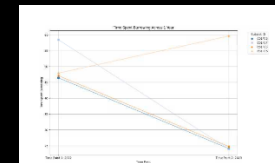


Validation (BORIS)

Keypoint MoSeq



Compare results



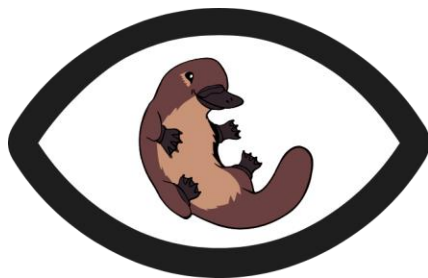


Manual classification was done for validation

BORIS application:

- **Validation** tool
- Classification of behavior (**Ethogram**)
- Requires classifying “burrowing” and “digging”
 - Burrowing – head disappears
 - Digging – shoulder and arm movement
- 12 videos analyzed on BORIS

Behavior	Code	Description
Digging	DG	Using the front paws to remove the rice while keeping the head up .
Burrowing	BG	Inserting the head into the rice and using the front paws to remove the rice while moving forward. Scratching and biting of the inner side of the rice container.
Sniffing	SG	Walking around the inside of the container while keeping the nose close to the rice surface.
Exploratory	EQ	Climbing over the container walls, using them as support to rest. Walking and sniffing while being outside of the enrichment area.
Digging outside	DO	Biting and scratching the container while being outside of the enrichment area.
ST	ST	Remaining static within the enrichment area





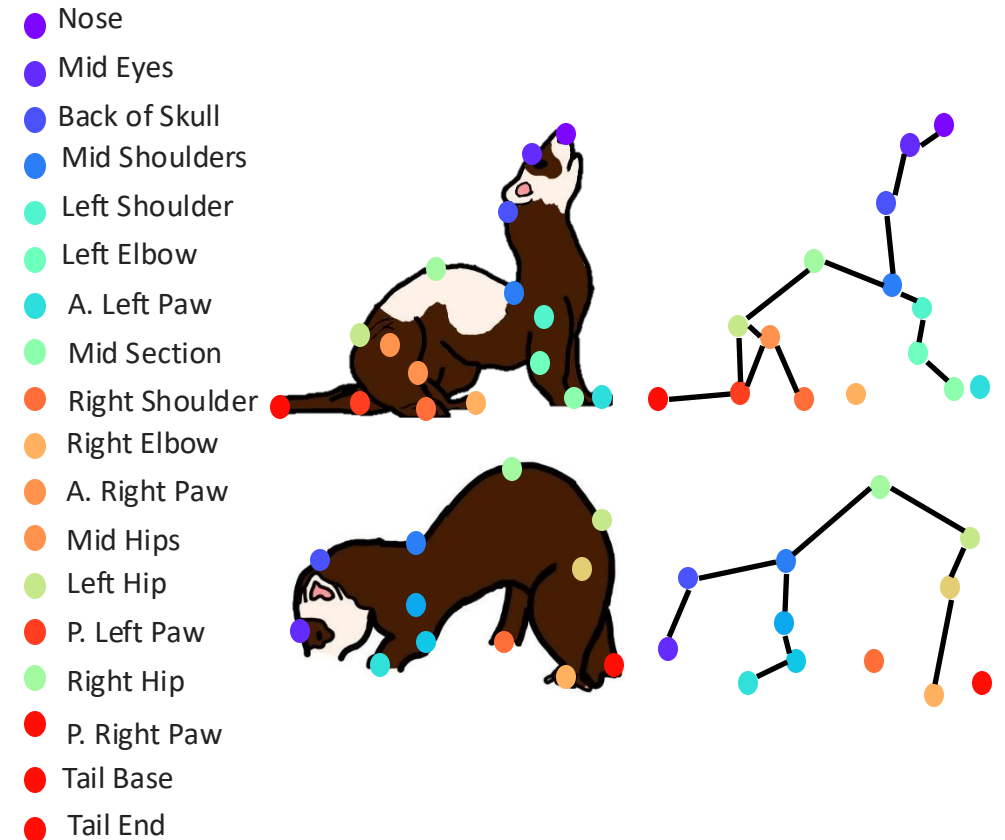
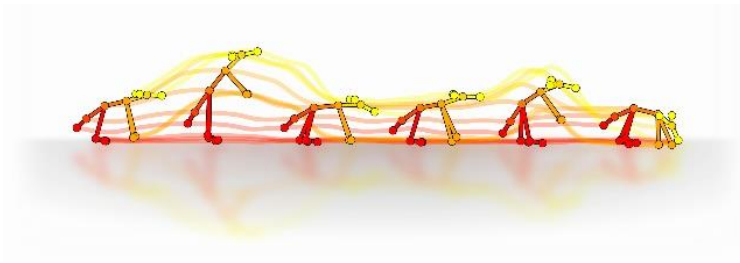
Automated behavior classification increases time efficiency and shows additional data

Deeplabcut parameters:

- 5% of the videos were used as **training videos**
- 20 frames were **manually** labeled

Keypoint Moseq:

- Discovers stereotypical movements
- Important for classification of more complex behaviors that involve motion

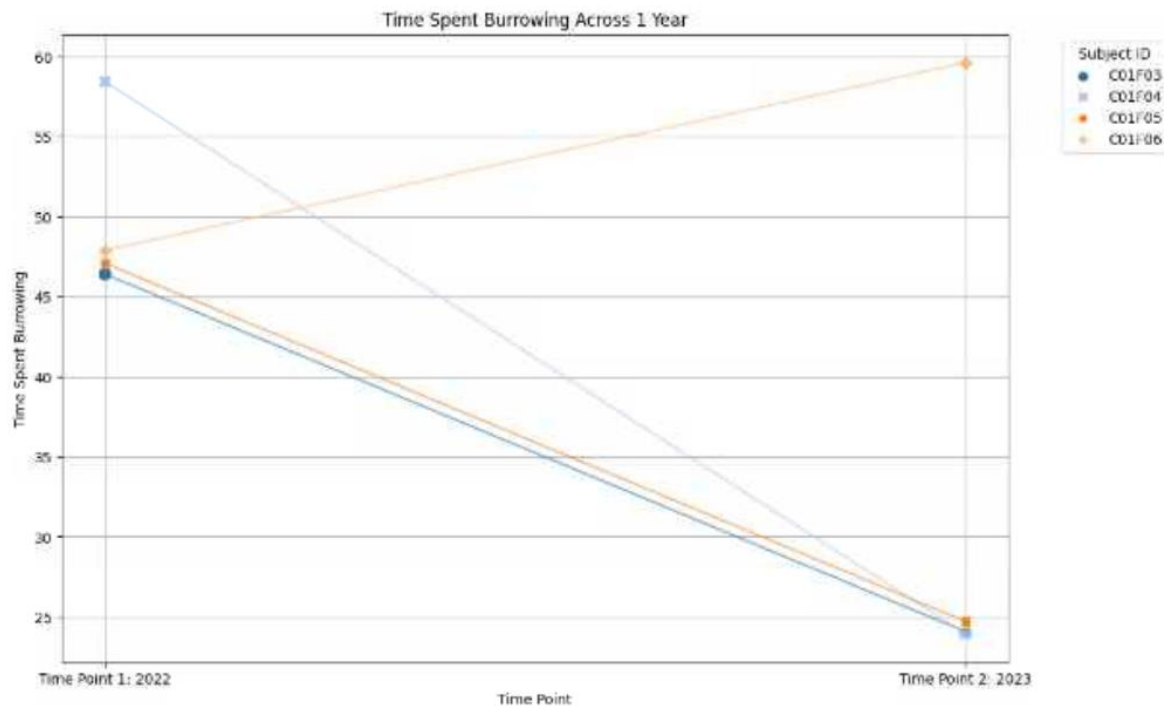


Frame labeling



Preliminary Results

BORIS burrowing analysis
2023-2024 (old ferrets)



Deeplabcut model needs
some refining

Training iterations:	Train error(px)	Test error(px)
100000	16.27	14.22



Future Directions

- Burrowing analysis
- Social interactions
- Sleep patterns
- MRI relationships





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Elizabeth B. Hutchinson
Principal Investigator

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**Thank you for your attention.
Questions?
Suggestions?**

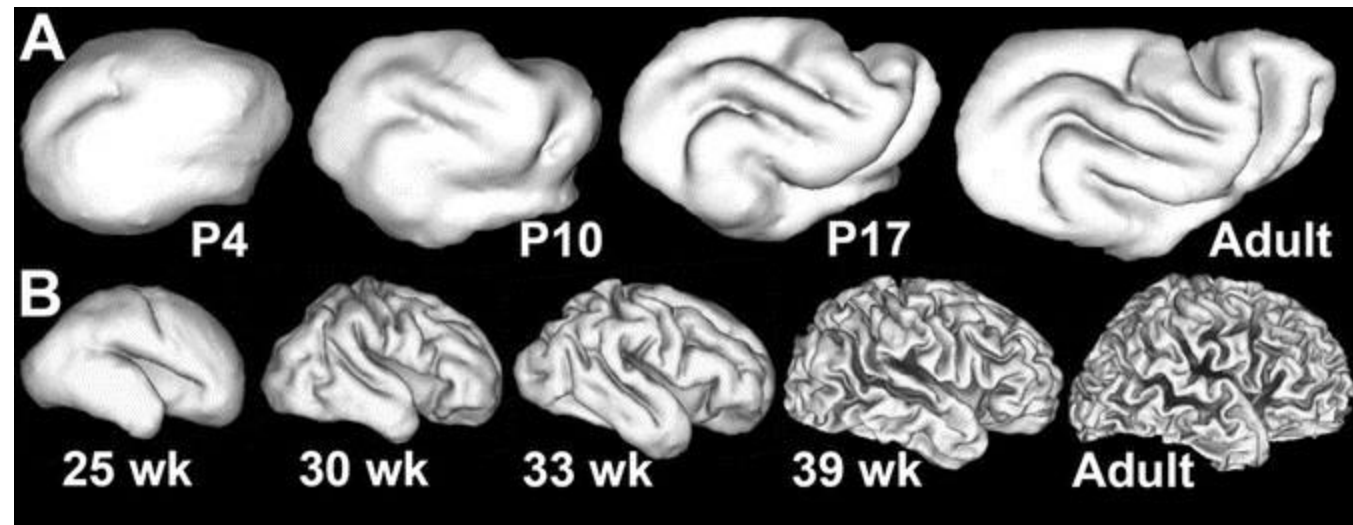


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Ferrets brain development





Deeplabcut markers video from rats

